

The Investigation for the Complexity of Convolutional Neural Network Model on Facial Emotion Recognition Performance

Xinmu Qi^{1, *}

¹The Department of Stony Brook, Anhui University, Hefei, 230000, China

*Corresponding author's e-mail address: R32014090@stu.ahu.edu.cn

Abstract. Emotions, being a fundamental facet of human experience, hold significant relevance in the realms of human communication and behavioural patterns. One of the most common applications of emotion detection is in Facial Emotion Recognition (FER), which relies on computer vision and pattern recognition techniques to analyse facial expressions and identify underlying emotional states. Two Convolutional Neural Network (CNN) models were constructed, including Visual Geometry Group (VGG) net and a self-defined CNN, which were trained on a dataset of human facial images with corresponding labels for different emotional states. The number of training epochs was selected to assure convergence and both models were trained using stochastic gradient descent or the Adam optimizer. The performance of both models was evaluated in terms of accuracy on a separate test set of images. The study also varied the number of parameters in the self-defined CNN model to retrain it on the FER dataset and evaluate its performance. The findings showed that increasing the number of parameters improved classification accuracy for the self-defined CNN model, demonstrating that models with more complex structure could lead to better classification performance. The study provides insights into the influence of the number of parameters on FER's performance, suggesting that optimal parameter choice significantly affects classification performance of CNN models. Overall, the results of this study provide valuable insights into the effectiveness of different CNN models on a specific task. Future research could build on these findings by exploring the influence of other hyperparameters and architecture choices on CNN performance.

Keywords: Facial Emotion Recognition, Convolutional Neural Network, Machine Learning

1. Introduction

Emotions, being a fundamental facet of human experience, hold significant relevance in the realms of human communication and behavioral patterns. In recent times, deep learning has emerged as the most popular way for researchers to develop automated systems capable of accomplishing classification tasks. In the classic classification task, emotion detection increasingly attracted much attention due to its potential applications in many domains, including marketing and advertising, customer service, mental health, education, and security. For instance, companies can utilize emotion detection to analyze customer sentiment and provide more targeted customer experiences, while mental health professionals can employ the technology to monitor patient emotions and provide personalized treatment.

Emotion detection, also known as affective computing, is a technology that uses artificial intelligence and machine learning algorithms to identify and based on information from physiological reactions, speech tonality, or facial expressions, assess the emotions of others. The goal of emotion detection is to recognize and classify emotions such as happiness, sadness, anger, fear, or surprise based on these signals. One of the most common applications of emotion detection is in Facial Emotion Recognition (FER), which relies on computer vision and pattern recognition techniques to analyze facial expressions and identify underlying emotional states.

FER has been extensively studied in the fields of computer vision, signal processing, and affective computing. Previous studies have focused on developing algorithms and models that can accurately identify and classify emotions based on facial expressions, as well as on evaluating the performance of these methods on benchmark datasets.

One of the recent works is implemented by Debnath et al. [1], who proposed a new FER model using a four-layer Convolutional Neural Network (CNN) called ConvNet. ConvNet detects seven specific emotions from image data: anger, disgust, fear, happiness, neutrality, sadness, and surprise. The authors used three datasets for training and testing, namely FER2013, JAFFE, and CK+. They achieved 96 % training accuracy and 91.01 % validation accuracy for the FER2013 dataset, and 92.05% and 98.13% accuracy for the JAFFE and CK+ datasets, respectively. Additionally, Khattak et al. [2], also proposed a powerful deep learning approach for identifying age and the type of facial expressions and the ranking of facial photo emotions based on a CNN model. They used the FER2013 dataset for training and testing their model. They achieved an accuracy of 95.65% for emotion recognition, 98.5% for age's, and 99.14% for gender's. Gannouni et al. [3] considered a different approach that used Electroencephalography (EEG) signals and CNNs for emotion detection. Different category schemes were defined using Recurrent Neural Network (RNN) and evaluated using the DEAP dataset. The average precision level was over 89%. Compared to the existing algorithms dealing with 9 emotions, the suggested method improved the degree of exactitude by 8%.

In this regard, this study aims to investigate the influence of the number of model parameters on the performance of FER. To compare the identical ability of different CNN in this dataset, the VGG net and a self-defined CNN were constructed. This study first prepared a dataset for FER with images of human faces in different emotional states, along with corresponding labels. Subsequently, both CNN models were trained on the aforementioned FER dataset, employing a consistent training protocol encompassing techniques such as optimization algorithm or the Adam optimizer. The number of times a machine learning algorithm processes the entire training dataset during its optimization process was appropriately set to ensure convergence of both models. Concurrently, the performance of both models on a separate test set of images was evaluated in terms of the classification accuracy. Next, this study varied the number of parameters and re-trained it on the FER dataset and evaluate the new model's performance on the same test set. Ultimately, this study reported the classification accuracy and compared the classification accuracy of the basic CNN model with varying numbers of parameters and the VGG model with a larger number of parameters and analyze the results to determine whether there is a significant difference in accuracy due to the variation in the number of parameters.

2. Method

2.1 Dataset description and preprocessing

In this study, the paper used the data collected from the Extended Cohn-Kanade dataset on Kaggle [4]. The primary objective of this study involved a supervised classification task, aiming to develop a model capable of predicting the emotion category associated with a given facial image. The input data consists of grayscale images with dimensions of 48×48 pixels, and the faces were subjected to an automated registration process, ensuring that the facial features were approximately centered and occupied a consistent proportion of space within each image. The output variable is a discrete emotion label from one of the seven categories (i.e. 0. "Angry", 1. "Disgust", 2. "Fear", 3. "Happy", 4. "Sad", 5.

"Surprise", 6. "Neutral"). The dataset contains a total of 32,298 examples, with 28,709 examples in the training set and 3,589 examples in the test set.

In terms of the image preprocessing, the main DataGenerator object with arguments was created, which can be categorized into four parts. Firstly, normalizing the dataset by dividing 255 to a uniform size. Secondly, using 20% of the data for validation to tune the hyperparameters and prevent overfitting during the training process. Thirdly, flipping, zooming the images randomly by 10% and rotating the images randomly by 20 degrees. Finally, using the above created DataGenerator object to create 2 generators: traingen and testgen. Additionally, Figure 1 is some visualizations of the augmented images from training data.

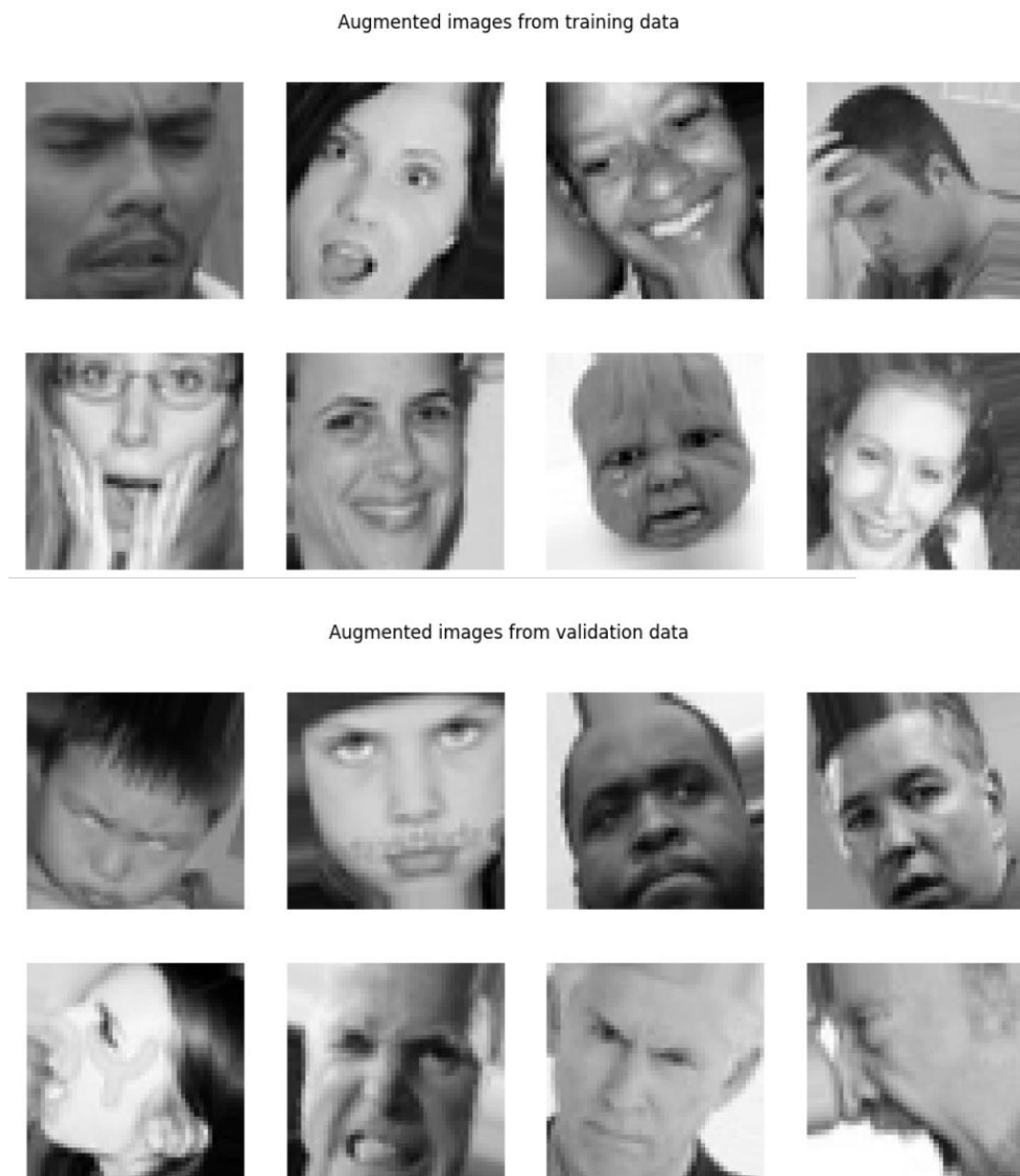


Figure 1. Visualizations of the augmented images from training data.

2.2 CNN-based emotion recognition model

This study employed prominent deep learning architectures, specifically CNNs and the VGG model, with certain modifications. CNN is a popular deep learning architecture that works well for image classification tasks [5-7]. It is designed to automatically learn relevant features from the input data using a combination of convolutional and pooling layers, which can learn spatial and temporal dependencies in the input data, making them ideal for image and video processing tasks. VGG is a profound convolution neural network architecture that was proposed by the researchers at the University of Oxford in 2014 [8]. The VGG model performed at the leading edge of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) dataset for object recognition, which includes over a million images across 1,000 classes [8]. The VGG model has been widely used in various computer vision applications such as object detection, image segmentation, and even in medical imaging. Despite being a simple architecture, it has proven to be very effective in achieving high accuracy on large-scale datasets.

The network employed in this study consists of six convolution layers and one completely connected layer in total, and each convolutional layer employs kernels that are 3×3 in size. The network first takes in 48×48 red and green and blue images taken from the dataset. Before being fed into the model, each image undergoes some preprocessing and augmentation techniques. The pixel values of the images are then normalized to range between 0 and 1. The network then takes the preprocessed image as input and processes it through its layers, traversing through the various layers. Following each group of convolution layers, a layer of max-pooling is incorporated to reduce the width and height of the feature map. Subsequently, a flatten layer is introduced to convert the output into a one-dimensional vector. Ultimately, the fully connected layers in the network produce a single probability score for the classification decision. The architectural configuration is visually depicted in Figure 2 and Figure 3.

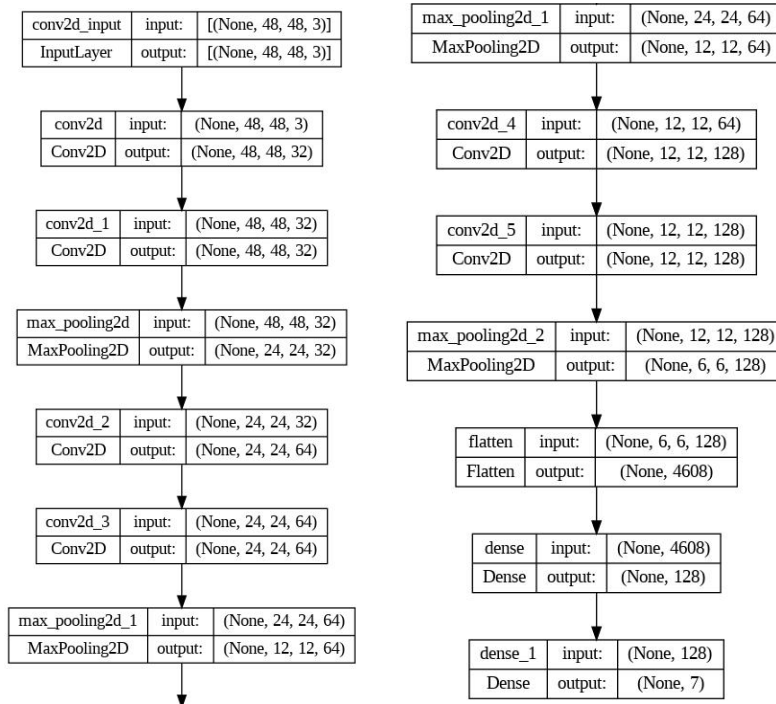


Figure 2. Model flow chart of the self-designed CNN.

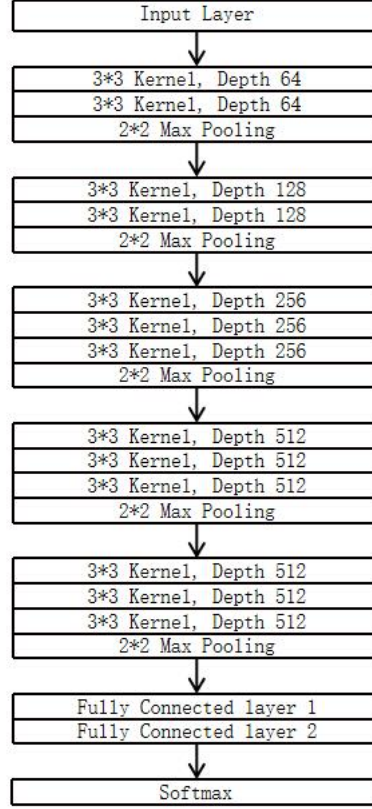


Figure 3. Model flow chart of the VGG

2.3 Implementation details

The categorical crossentropy loss function which is a commonly used loss function in machine learning for multi-class classification problems where the output labels are one-hot encoded was employed. The model calculates the dissimilarity between the predicted probability distribution and the true probability distribution of the target class. The formula for categorical crossentropy loss is as follows:

$$L = - \sum (y * \log(y_{pred})) \quad (1)$$

Where L , y and y_{pred} are categorical crossentropy loss, one-hot encoded true label vector and predicted probability vector for each class, respectively.

The evaluation metrics, the accuracy metric, will be used to assess the performance of the model. Accuracy is a popular evaluation criterion that gauges the proportion of correct predictions made by the model among all the predictions generated. It provides insights into how well the model can classify examples from different classes and how much it can generalize to unseen data.

The optimizer is a key component in training neural network models. It determines how the parameters of the model that are learned during training, including both the weights and biases are updated during training to minimize the loss function. Stochastic gradient descent (SGD) is a popular optimization algorithm used to update the weights of a neural network during training [9, 10]. The SGD optimizer is configured with a rate of learning with the value of 0.001 and a momentum value of 0.9. Additionally, the epoch number is set to 20.

3. Results and discussion

3.1. The performance of the model

A series of experiments was conducted to evaluate the impact of adjusting epoch and batch size, as well as incorporating the VGG-16 model, on the performance of the dataset. The experiments demonstrated that the self-designed CNN model (epoch=20, batch size=32) has the best performance as shown in Table 1, and it turned out that in the training, the proposed model had a loss of 1.0746 and accuracy of 0.5965 after 20 epochs of training while it had loss of 1.1428 and accuracy of 0.5668 in the validation set. When the model applied to the test set, the model has a commendable generalization ability, yielding an accuracy rate of 0.5831.

Table 1. The performance of different models with different parameter.

Model	Performance				
	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy	Testing Accuracy
Self-designed CNN1 (epoch=10, batch size = 32)	1.1506	0.5664	1.2027	0.5440	0.5454
Self-designed CNN2 (epoch=20, batch size = 32)	1.0746	0.5965	1.1428	0.5668	0.5831
Self-designed CNN3 (epoch=10, batch size = 128)	1.1724	0.5604	1.2243	0.5387	0.5402
VGG-16 (epoch = 10, batch size = 32)	1.1487	0.5698	1.1998	0.5479	0.5483

Overall, for the VGG-16 model, training for 10 epochs with a batch size of 32 produced a validation accuracy of 0.5479, which is slightly lower than the best performance achieved by the self-designed CNN model. In detail, these results suggest that increasing the number of epochs while maintaining a reasonable batch size can improve the performance of the self-designed CNN model in

this study. Meanwhile, the VGG-16 model may require more than 10 epochs to achieve better performance on this specific task.

3.2. Discussion

The results indicate that the self-designed CNN models outperformed the VGG-16 model in terms of validation accuracy, training the model over 20 epochs, where each epoch involves iterating over the entire dataset in batches of 32 examples. This discovery aligns with previous findings, indicating that the results are in agreement with what has been observed before. This study suggest that CNN models designed by self can achieve high accuracy on various tasks, especially when combined with effective hyperparameter tuning.

The generalizability of the findings presented in this study is contingent upon the specific dataset and task employed. Consequently, it is crucial to exercise caution when applying these results to different tasks or datasets, as variations in data characteristics may influence the outcomes. However, the results underscore that researchers should consider exploring their own self-designed CNN models for similar tasks and compare their performance against established models like VGG-16. In terms of the impact of epoch and batch size, the results suggested that increasing the number of epochs while maintaining a reasonable batch size can improve the performance of the self-designed CNN model. However, increasing the batch size while keeping the same number of epochs did not lead to improved performance. It's worth noting that these results are consistent with previous research, which has shown that optimizing the number of epochs and batch size is an essential part of building an accurate CNN model.

Overall, the results of this study provide valuable insights into the effectiveness of different CNN models on a specific task. Future research could build on these findings by exploring the influence of other hyperparameters and architecture choices on CNN performance. However, there are still some aspects of this study that can be improved : 1) Comparing with more models: The study only compares the performance of two CNN models - a self-designed CNN and VGG-16. Including additional models in the comparison, such as ResNet or Inception, would provide a more comprehensive understanding of the effectiveness of different CNN architectures on this task. 2) Fine-tuning hyperparameters: Although the study varied the number of epochs and batch size, it's possible that other hyperparameters, such as learning rate or weight decay, could be further fine-tuned to improve the performance of the CNN models. A more exhaustive hyperparameter search could lead to even higher accuracy scores for both models.

4. Conclusion

In conclusion, this study aimed to develop a model capable of predicting the emotion category associated with a given facial image using prominent deep learning architectures, specifically self-designed CNNs and the VGG model, with certain modifications of some parameters. The primary objective of the study was to achieve a supervised classification task that accurately predicts emotions from facial images. Through experimentation and analysis, the study found that modifying certain parameters in the self-designed CNN and VGG models can have a significant impact on their performance in predicting emotions from facial images. The results demonstrated that both models achieved high accuracy rates in predicting emotions from images depicting the countenance of a person, showcasing their facial features and expressions, with the CNNs achieving slightly enhanced precision than the VGG model. These findings suggest that deep learning models can be highly effective in accurately predicting emotions from facial images and have important implications for the development of emotion recognition systems in various fields, including psychology, marketing, and security.

5. References

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